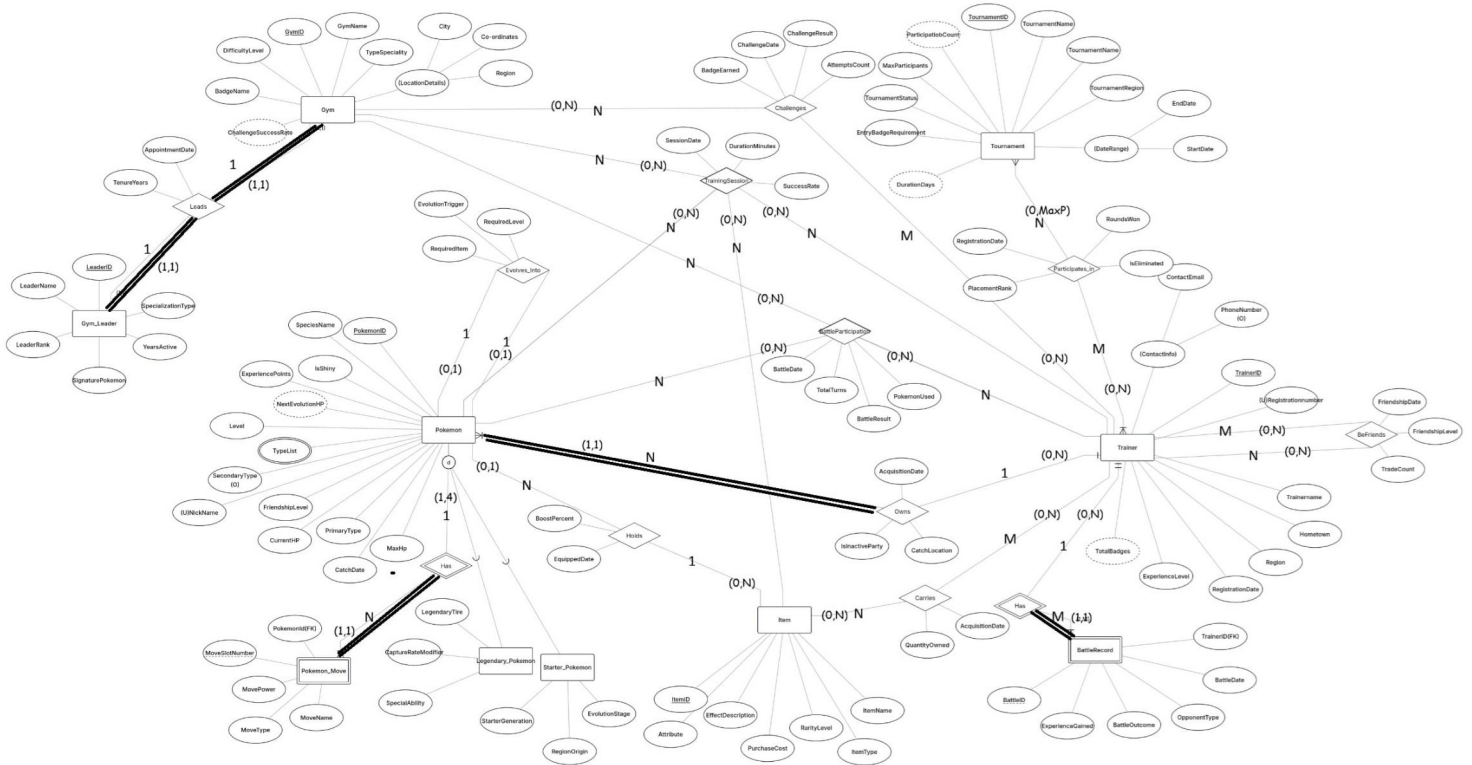


Karasuno - phase-2



Minor Edits from Phase 1:

No significant changes were made to the original requirements. All entities, relationships, attributes, and constraints from Phase 1 have been faithfully represented in the ER diagram using standard notation and ERDPlus tool capabilities.

Design Quality Analysis

5.1 Redundancy Analysis

Identified Redundancies:

1. **WinRate (TRAINER)** - Derivable from BATTLE_RECORD outcomes
2. **TotalBadges (TRAINER)** - Countable from CHALLENGES where BadgeEarned = TRUE
3. **ChallengeSuccessRate (GYM)** - Computable from CHALLENGES data
4. **ParticipationCount (TOURNAMENT)** - Countable from PARTICIPATES_IN records
5. **NextEvolutionXP (POKEMON)** - Calculable from Level and ExperiencePoints
6. **DurationDays (TOURNAMENT)** - Computable from StartDate and EndDate

Problems Caused:

- Data inconsistency when base data updates but derived values don't recalculate

- Wasted storage space
- Conflicting values between computed and stored data

5.2 Update Anomaly

Yes, update anomalies exist.

Example 1: When a trainer earns a new badge through CHALLENGES:

- Must update BadgeEarned in CHALLENGES relationship
- Must update TotalBadges in TRAINER entity
- Must update BadgeCollection multi-valued attribute in TRAINER
- Must recalculate WinRate if battle records change

Failure to update all locations creates inconsistent badge information.

Example 2: When Pokemon stats change (level up):

- Must update Level in POKEMON
- Must update ExperiencePoints in POKEMON
- Must recalculate NextEvolutionXP
- If battle participation is recorded with stats, those become outdated

5.3 Insertion Anomaly

Yes, insertion anomalies exist.

Scenario 1: Cannot insert a new GYM without a GYM_LEADER because the LEADS relationship has total participation on both sides (1:1, both double lines). A newly constructed gym cannot exist in the database until a leader is appointed, even if the building is complete.

Scenario 2: Cannot insert a POKEMON_MOVE without an existing POKEMON because it's a weak entity with total participation in the identifying relationship. Cannot pre-define moves before catching the Pokemon.

Scenario 3: Cannot insert a BATTLE_RECORD without a TRAINER because it's a weak entity totally dependent on TRAINER. Cannot record NPC or wild Pokemon battles without a trainer entity.

5.4 Deletion Anomaly

Yes, deletion anomalies exist.

Example 1: Deleting a TRAINER who owns POKEMON triggers CASCADE deletion (per stated referential integrity rules):

- Loses all owned POKEMON records permanently
- Loses all BATTLE_RECORD history for that trainer
- Loses all POKEMON_MOVE data for those Pokemon
- Loses CHALLENGES history (gym battle records)
- Loses PARTICIPATES_IN tournament participation data

This causes permanent loss of valuable battle statistics and historical competitive data.

Example 2: Deleting a TOURNAMENT removes all PARTICIPATES_IN records:

- Loses all participant PlacementRank data
- Loses RoundsWon statistics
- Cannot analyze trainer performance history across tournaments
- Historical tournament data becomes unrecoverable

Example 3: Deleting a GYM_LEADER who LEADS a gym violates the referential integrity RESTRICT rule, preventing deletion even if legitimate (retirement, transfer). This creates operational difficulties when leaders need to be removed from the system.

Example 4: Deleting a POKEMON involved in EVOLVES_INTO relationships:

- Loses evolution chain information
- Breaks historical tracking of which Pokemon evolved from what
- Cannot reconstruct evolution history for analytics